

3.3.1 Cooking of food and heat transfer

3.3.1.1 Why food is cooked and how heat is transferred to food

Content	Students must know and understand	Suggested application and food preparation skills
- the reasons why food is cooked - the different methods of heat transfer.	Food is cooked to: - make food safe to eat - develop flavours - improve texture - improve shelf life - give variety in the diet. How preparation and cooking affect the appearance, colour, flavour, texture, smell and overall palatability of food. How heat is transferred to food through: - conduction - convection - radiation.	For sauce making: - how conduction and convection work to cook a sauce and the need for agitation (S6) - how radiation works using the grill for a range of foods such as vegetables, meat, fish or alternatives such as halloumi, seeds and nuts, to char, toast and grill (S4).

3.3.1.2 Selecting appropriate cooking methods

Content	Students must know and understand	Suggested application and food preparation skills
Selection of appropriate preparation, cooking methods and times to achieve desired characteristics.	• how the selection of appropriate preparation and cooking methods can conserve or modify nutritive value or improve palatability: • water based: steaming, boiling, simmering, blanching, poaching, braising • dry methods: baking, roasting, grilling, dry frying • fat based: shallow frying, stir fry • how preparation and cooking affect the appearance, colour, flavour, texture, smell and overall palatability of food eg the use of marinades to denature protein.	• Using the oven for baking, roasting, braising, casseroles and/or tagines (S4). • Dry heat and fat based methods using the hob; dry frying, shallow frying and stir frying (S6). • Use of the microwave oven (S5). • Water, dry heat and fat based cooking methods using the hob – to conserve nutritive value eg steaming, stir frying (S6). • General practical skills – judge and modify sensory properties – awareness of the effect of preparation and cooking on the sensory characteristics of food – appearance, colour, flavour, texture, taste and season adding herbs, spices etc. Use browning and glazing to change texture and flavour. Improve aesthetic qualities of foods by garnishing and decorating (S1). • The use of marinades to tenderise and flavour meats and alternatives (S9). • The boiling of vegetables to alter texture (S6).